

AI Smart Parking System - User Guide & Classification

Overview

An AI Smart Parking System automates vehicle entry, number plate recognition, parking allocation, monitoring, and exit processing using AI and computer vision.

How to Use

1. Start the backend server (FastAPI/Python).
2. Start the React frontend.
3. Open the web application in your browser.
4. Allow camera access.
5. Drive the vehicle in front of the camera.
6. The AI detects the vehicle and number plate.
7. OCR extracts the registration number.
8. The system checks if the vehicle is registered.
9. An available parking slot is assigned automatically.
10. On exit, the plate is recognized again and the slot is released.

AI Classification

- Vehicle Detection: YOLOv8 detects cars, motorcycles, buses, trucks.
- Number Plate Detection: AI locates the license plate.
- OCR: Reads characters from the plate image.
- Slot Classification: Empty vs Occupied parking spaces.
- Entry/Exit Classification: Determines whether a vehicle is entering or leaving.

System Flow

Camera → YOLO Vehicle Detection → Plate Detection → OCR → Database Verification → Slot Assignment → Parking Database Update → Exit Detection → Slot Release.

Recommended Technologies

Frontend: React

Backend: FastAPI (Python)

AI: YOLOv8

OCR: EasyOCR or PaddleOCR

Database: MySQL or PostgreSQL